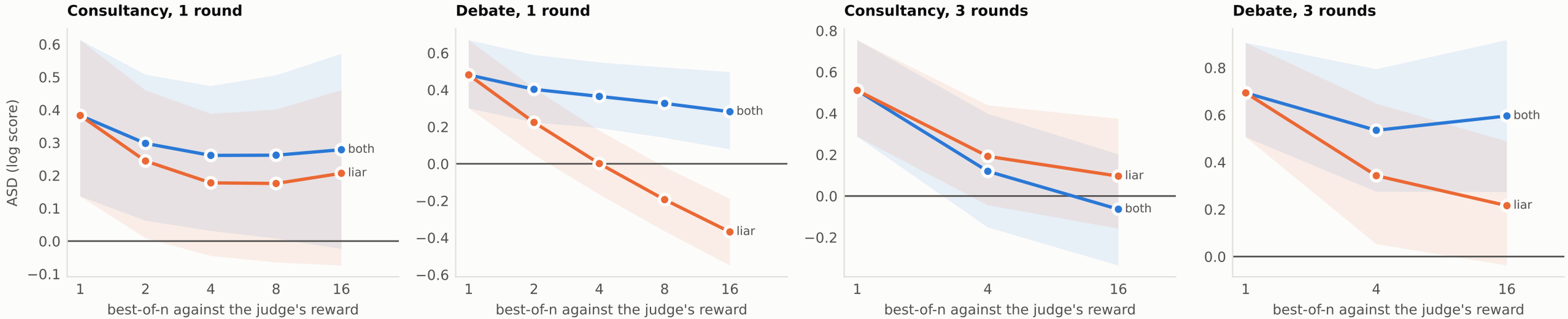
Optimizing only the liar erodes the incentive to be honest; three rounds and symmetric optimization keep it

Blue: both sides optimized. Orange: only the liar optimized. gpt-5-mini; held-out judging (selection and evaluation on independent judge samples); 3 judges pooled; 95% item-bootstrap bands.